

LSM 203 FORECASTING AND DEMAND MANAGEMENT

PRACTICE QUESTIONS

1. Demand forecasting is best described as the process of:
 - A. Predicting supply capacity
 - B. Estimating future customer demand
 - C. Determining production costs
 - D. Setting selling prices
2. Which of the following is NOT a component of demand?
 - A. Trend
 - B. Seasonality
 - C. Randomness
 - D. Inflation
3. Demand forecasting primarily supports:
 - A. Employee appraisal
 - B. Strategic and operational planning
 - C. Financial auditing
 - D. Product branding
4. Which demand pattern shows long-term movement?
 - A. Seasonal
 - B. Cyclical
 - C. Trend
 - D. Random
5. Seasonal demand is best associated with:
 - A. Economic recessions
 - B. Weather and festive periods
 - C. Random fluctuations
 - D. Pricing errors
6. A forecasting method based on expert opinion is:
 - A. Moving average
 - B. Regression analysis
 - C. Delphi method
 - D. Exponential smoothing
7. Qualitative forecasting is most suitable when:
 - A. Historical data is abundant
 - B. Demand is stable
 - C. Historical data is limited
 - D. Demand is predictable
8. Which of the following is a quantitative forecasting method?
 - A. Market survey
 - B. Sales force opinion

- C. Time series analysis
 - D. Jury of executives
9. A simple moving average forecast relies on:
- A. The most recent observation only
 - B. Weighted historical data
 - C. Average of past observations
 - D. Regression coefficients
10. Increasing the number of periods in a moving average will:
- A. Increase forecast error
 - B. Make forecasts more responsive
 - C. Smooth demand fluctuations
 - D. Eliminate seasonality
11. Exponential smoothing assigns:
- A. Equal weights to all data
 - B. Higher weight to recent data
 - C. No weight to past data
 - D. Random weights
12. The smoothing constant (α) in exponential smoothing must lie between:
- A. 0 and 10
 - B. 1 and 5
 - C. 0 and 1
 - D. -1 and 1
13. A higher value of α makes the forecast:
- A. Less responsive
 - B. More responsive to changes
 - C. More stable
 - D. Independent of demand
14. Which forecasting method captures trend explicitly?
- A. Simple moving average
 - B. Naïve method
 - C. Trend projection
 - D. Delphi method
15. Regression forecasting establishes a relationship between:
- A. Demand and cost
 - B. Dependent and independent variables
 - C. Demand and inventory
 - D. Trend and seasonality
16. Which of the following measures forecast accuracy?
- A. EOQ
 - B. MAD
 - C. Safety stock
 - D. Reorder level

17. Mean Absolute Deviation (MAD) measures:
 - A. Squared forecast errors
 - B. Percentage error
 - C. Average absolute error
 - D. Bias direction
18. Mean Squared Error (MSE) penalizes:
 - A. Small errors
 - B. Large errors more heavily
 - C. Random errors only
 - D. Bias only
19. Forecast bias indicates:
 - A. Randomness in demand
 - B. Consistent over- or under-forecasting
 - C. Seasonal variation
 - D. Demand stability
20. Tracking signal is used to:
 - A. Identify seasonality
 - B. Monitor forecast bias
 - C. Reduce inventory
 - D. Improve pricing
21. Demand management focuses on:
 - A. Eliminating uncertainty
 - B. Influencing demand patterns
 - C. Increasing production
 - D. Minimizing labour
22. Demand shaping tools include:
 - A. Capacity expansion
 - B. Promotions and pricing
 - C. Outsourcing
 - D. Automation
23. Which strategy is used to reduce peak demand?
 - A. Price increase
 - B. Off-peak discounts
 - C. Stockpiling
 - D. Advertising
24. Collaborative forecasting involves:
 - A. Internal departments only
 - B. Customers and suppliers
 - C. Financial institutions
 - D. Regulators
25. CPFR stands for:
 - A. Collaborative Production Forecast Review

- B. Collaborative Planning, Forecasting and Replenishment
 - C. Coordinated Product Flow Regulation
 - D. Centralized Planning and Forecast Rule
26. Demand variability is often amplified by:
- A. Accurate forecasting
 - B. Information sharing
 - C. Bullwhip effect
 - D. Stable demand
27. The bullwhip effect results from:
- A. High inventory levels
 - B. Demand information distortion
 - C. Stable pricing
 - D. Short lead times
28. Which sector relies heavily on demand forecasting?
- A. Transportation
 - B. Logistics and retail
 - C. Education
 - D. Tourism
29. Poor forecasting often leads to:
- A. Lower costs
 - B. Excess inventory or stockouts
 - C. Higher customer satisfaction
 - D. Stable supply chains
30. Demand forecasting is critical for:
- A. Facility location
 - B. Inventory control
 - C. Capacity planning
 - D. All of the above
31. Life-cycle forecasting is especially useful for:
- A. Mature products
 - B. Declining products
 - C. New products
 - D. Staple goods
32. Which stage of product life cycle has the highest uncertainty?
- A. Growth
 - B. Maturity
 - C. Introduction
 - D. Decline
33. Forecasting horizon refers to:
- A. Forecast accuracy
 - B. Time period covered

- C. Demand volume
 - D. Data frequency
34. Short-term forecasts are mainly used for:
- A. Strategic planning
 - B. Capacity expansion
 - C. Operational decisions
 - D. Market entry
35. Long-term forecasts support:
- A. Daily scheduling
 - B. Inventory counting
 - C. Strategic planning
 - D. Order picking
36. Which method is easiest but least accurate?
- A. Regression
 - B. Naïve method
 - C. Exponential smoothing
 - D. Trend analysis
37. Demand planning integrates forecasting with
- A. Finance only
 - B. Production and inventory decisions
 - C. Marketing only
 - D. Transportation only
38. Forecast error is the difference between
- A. Forecast and budget
 - B. Actual demand and forecast
 - C. Inventory and sales
 - D. Capacity and demand
39. Forecast accuracy improves with
- A. More assumptions
 - B. Better data quality
 - C. Higher α
 - D. Longer lead time
40. Which factor affects forecast reliability?
- A. Demand stability
 - B. Data availability
 - C. Forecast horizon
 - D. All of the above
41. Demand management aims to:
- A. Eliminate demand
 - B. Match demand with supply
 - C. Increase forecasting error
 - D. Reduce customer choice

42. Which of the following reduces demand uncertainty?
- A. Information sharing
 - B. Long lead times
 - C. Price fluctuations
 - D. Poor coordination
43. Public sector demand forecasting is challenging due to:
- A. Stable demand
 - B. Political and budget constraints
 - C. Excess data
 - D. Overcapacity
44. Transport demand forecasting helps in:
- A. Route planning
 - B. Fleet sizing
 - C. Infrastructure planning
 - D. All of the above
45. Forecast aggregation generally:
- A. Increases error
 - B. Reduces variability
 - C. Eliminates uncertainty
 - D. Distorts demand
46. Which method is best for stable demand?
- A. Moving average
 - B. Trend projection
 - C. Regression
 - D. Delphi
47. Which forecasting approach relies on historical patterns?
- A. Qualitative
 - B. Judgmental
 - C. Time series
 - D. Intuitive
48. Demand forecasting errors cannot be:
- A. Measured
 - B. Eliminated completely
 - C. Reduced
 - D. Monitored
49. A good forecasting system should be:
- A. Complex
 - B. Expensive
 - C. Simple and reliable
 - D. Fully automated only
50. The main objective of forecasting is to:
- A. Predict the future perfectly

- B. Reduce uncertainty in decision-making
- C. Eliminate risk
- D. Increase production

1. Explain five key objectives of demand forecasting in logistics and supply chain management.
2. Given the following demand data:
Month 1: 350; Month 2: 370; Month 3: 390. Using weights of 0.2, 0.3, and 0.5, respectively, forecast demand for Month 4.
3. Briefly discuss five components of demand and how each affects forecasting accuracy.
4. Discuss five factors influencing the choice of a forecasting method.
5. Explain demand management and state five demand shaping tools
6. Discuss role of forecasting in supply chain operation
7. A logistics company observes the following relationship between advertising spend (X) and monthly demand (Y). X (₦): 40,000, 50,000, 60,000, 70,000
Y (units): 350, 410, 460, 500
Estimate the regression equation and forecast demand when advertising spend is ₦150,000.